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Problem 1

Show that the value of ϕ in vacuum at any point is equal to the average of ϕ over the surface of any sphere centered at that point.

HINT: Use Green's theorem with appropriate φ and ψ .

Let $\varphi = \phi$, the electric potential. If we represent ϕ (electric potential) as $\phi = G(r, r') + F(r, r')$, then let $\psi = G(r, r')$.

In a vacuum, there is no charge density, so $\nabla^2 \phi = 0$ unless we have a free particle at the origin.

$$\nabla^2 G = 4\pi\delta^3(x - x')$$

$$G = \frac{1}{|\vec{r} - \vec{r}'|}, \text{ where } \vec{r}' = \vec{0}.$$

We have the boundary condition that at some radius R of a sphere we are enforcing some $\phi_0(\theta, \phi)$.

We want to show that

$$\frac{1}{4\pi R^2} \oint \phi(R, \theta, \phi) \cdot da = \phi(r = 0)$$

We have green's function

$$\begin{aligned} \int \varphi \nabla^2 \psi - \psi \nabla^2 \varphi &= \oint \varphi \vec{\nabla} \psi - \psi \vec{\nabla} \varphi \\ \int \phi(4\pi\delta^3(\vec{r})) - \frac{1}{\vec{r}} 0 dv &= \oint \varphi \vec{\nabla} \psi - \psi \vec{\nabla} \varphi \cdot da \\ 4\pi\phi(\vec{r}) &= \oint \phi \frac{d}{dr} \left(\frac{1}{|r|} \right) - 0 \\ 4\pi\phi(\vec{r}) &= \oint \frac{\phi}{r^2} \\ \phi(\vec{r}) &= \frac{1}{4\pi R^2} \oint \phi \cdot da \end{aligned}$$

QED

I HATE IT - we are in a vacuum and don't have charge, but say that we have charge for the delta function thing (charge at the origin), but also say that we don't have any charge because $\psi \vec{\nabla} \phi = 0$ from a gauss's law argument

Problem 2

- Find ϕ for a grounded conducting sphere in a uniform electric field E_0 using the method of images.
- Find the surface charge density on the sphere and show that the total charge is zero.

We can replace the electric field with a dipole, placing q and $-q$ a distance a away from the origin. We can place imaginary charges inside the sphere at a distance a' from the origin.

The field given by the two points is

$$E = \frac{2q}{4\pi\epsilon_0 r^2}$$

To mimic the constant electric field, we take $r \rightarrow \infty$.

The field from the imaginary charges and the external has to be zero at the surface, so the imaginary charges have to have the same field at that radius

$$\frac{q'}{4\pi\epsilon_0 |R - a'|} + \frac{q'}{4\pi\epsilon_0 |R + a'|} = -\phi$$

We can solve for a' . Each pair has to cancel with its corresponding opposite charge.

$$\begin{aligned}\frac{q'}{|R - a'|} &= -\frac{q}{|R - a|} \\ q' |R - a| &= q |R - a'| \\ \frac{q'}{q} &= \left(\frac{|R - a'|}{|R - a|} \right) \\ \frac{q'}{q} &= \left(\frac{\sqrt{R^2 + a'^2 - 2Ra' \cos \theta'}}{\sqrt{R^2 + a^2 - 2Ra \cos \theta}} \right)\end{aligned}$$

We can pull out R from the q' term, and a from the q term

$$\frac{q'}{q} = \frac{R}{a} \frac{\sqrt{1 + \frac{a'^2}{R^2} - \frac{2a'}{R} \cos \theta'}}{\sqrt{1 + \frac{R^2}{a^2} - \frac{2R}{a} \cos \theta}}$$

We need

$$0 = \frac{q'}{R \sqrt{1 + \frac{a'^2}{R^2} - \frac{2a'}{R} \cos \theta'}} - \frac{q}{a \sqrt{1 + \frac{R^2}{a^2} - \frac{2R}{a} \cos \theta}}$$

These have to be the same, so we have

$$\frac{q'}{R^2} = \frac{q}{a}$$

to get the numerators to match, and

$$\frac{a'}{R} = \frac{R}{a}$$

We have now put conditions for the positions of the imaginary charges.

The net field from the charges and the imaginary charges is now

$$\begin{aligned}\phi &= \frac{2q}{4\pi\epsilon_0 a} - \frac{2q'}{r\pi\epsilon_0 a'} \\ \phi &= \frac{2q}{4\pi\epsilon_0 a} - \frac{2q}{4\pi\epsilon_0}\end{aligned}$$

These satisfy the field being 0 at radius R !

Problem 3

Consider a hollow conducting grounded sphere of radius b with a concentric ring of charge inside it given by

$$\rho(\vec{x}') = \frac{Q}{2\pi a^2} \delta(r' - a) \delta(\cos \theta').$$

Find ϕ everywhere inside the sphere.

HINT: For this, you need the Dirichlet Green's function G_D for a region bounded by two spheres of radii R_1 and R_2 , where $R_1 < R_2$:

$$G_D(\vec{x}, \vec{x}') = 4\pi \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{Y_{\ell m}^*(\theta', \varphi') Y_{\ell m}(\theta, \varphi)}{(2\ell+1) \left(1 - \left(\frac{R_1}{R_2}\right)^{2\ell+1}\right)} \left(r_{<}^{\ell} - \frac{R_1^{2\ell+1}}{r_{<}^{\ell+1}}\right) \left(\frac{1}{r_{>}^{\ell+1}} - \frac{r_{>}^{\ell}}{R_2^{2\ell+1}}\right)$$

with $R_1 \rightarrow 0$ and $R_2 = b$.

Verify the $b \rightarrow \infty$ limit with the result from class. You may leave your answer in the form of an infinite sum.

Problem 4

Consider a hollow conducting grounded sphere of radius b with a uniform line charge of total charge Q located on the z -axis between the north and south poles.

(a) Show that

$$\rho(\vec{x}') = \frac{Q}{2b} \frac{1}{2\pi r'^2} [\delta(\cos \theta' - 1) + \delta(\cos \theta' + 1)].$$

(b) Find ϕ everywhere inside the sphere.

HINT: For this, you need the Dirichlet Green's function G_D for a region bounded by two spheres of radii a and b , where $a < b$:

$$G_D(\vec{x}, \vec{x}') = 4\pi \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{Y_{\ell m}^*(\theta', \varphi') Y_{\ell m}(\theta, \varphi)}{(2\ell+1) \left(1 - \left(\frac{a}{b}\right)^{2\ell+1}\right)} \left(r_{<}^{\ell} - \frac{a^{2\ell+1}}{r_{<}^{\ell+1}} \right) \left(\frac{1}{r_{>}^{\ell+1}} - \frac{r_{>}^{\ell}}{b^{2\ell+1}} \right).$$

You will also need

$$\frac{d}{dy} (x^y) = x^y \ln x, \quad P_{\ell}(1) = 1, \quad P_{\ell}(-1) = (-1)^{\ell}.$$

(c) Find the surface charge density on the sphere.